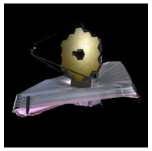

3696 - A population of hidden tidal disruptions in the local universe: Revealing the energetics of the most luminous infrared transients with JWST

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
WTP14adbjsh-MRS				
	1		MIRI Medium Resolution Spectroscopy	(1) NGC-7392
WTP14adbjsh-MRS-BKG				
	2		MIRI Medium Resolution Spectroscopy	(5) NGC7392-BACKGROUND
WTP18aajkmk-MRS				
	3		MIRI Medium Resolution Spectroscopy	(2) MCG-01-07-028
WTP18aajkmk-MRS-BKG				
	4		MIRI Medium Resolution Spectroscopy	(6) MCG-01-07-028-BACKGROUND

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
WTP18aampwj-MRS				
	5		MIRI Medium Resolution Spectroscopy	(3) 2MASX-J01464244+3230295
WTP18aampwj-MRS-BKG				
	6		MIRI Medium Resolution Spectroscopy	(7) KUG0143-BACKGROUND
WTP17aamzew-MRS				
	7		MIRI Medium Resolution Spectroscopy	(4) NGC-2981
WTP17aamzew-MRS-BKG				
	8		MIRI Medium Resolution Spectroscopy	(8) NGC2981-BACKGROUND

ABSTRACT

The physics of supermassive black holes (SMBHs) underpins all areas of astrophysics. It has long been suggested that SMBHs at the lowest masses ($\lesssim 10^6$\,M$_\odot$) grow substantially by the repeated tidal disruption of nearby stars; yet observational tests remain challenging. While such tidal disruption events (TDEs) are now routinely discovered in optical/X-ray surveys, there is growing evidence that they reveal only a fraction of the population where the SMBH line of sight is unobscured. Combined with most of the accretion luminosity emerging in the opaque extreme ultraviolet bands, both the bolometric radiated energy (tracing the accreted mass) and volumetric rates of TDEs remain highly uncertain. The mid-infrared (MIR) bands, acting as a bolometer for TDE radiation via reprocessed emission from surrounding dust, offer a unique opportunity to alleviate both these problems. By mining public NEOWISE images using state-of-the-art image subtraction methods, we have identified a sample of previously unknown luminous (> 10^{43} erg/s) MIR TDE candidates in nearby galaxies (< 200 Mpc). We propose MIRI/MRS spectroscopy of four sources showing diverse signs supporting their TDE origin in multi-wavelength follow-up. These first-of-their kind MIR observations of TDEs, possible only with JWST, will reveal i) the composition, temperature and geometry of dust heated in TDEs to measure the flare bolometric luminosity and ii) unmistakable evidence for an awakened SMBH via high ionization emission lines. Together, this proposal will provide a stepping stone to the characterization of TDEs in the MIR, and assess their contribution in forming the lightest SMBHs in the universe.

OBSERVING DESCRIPTION

The proposal will carry our MRS IFU spectroscopy of four tidal disruption events (TDEs) identified via their bright mid-infrared echoes in WISE data. We will use all the channels and grating settings to cover the full wavelength range from 5 - 28 microns. The observations will be carried out using the standard 4-point dither pattern for point sources. Since the background stellar light will fill the MRS field of view, we will obtain a single backgroung sky exposure matching the groups and integrations of the science observations, immediately after the target observations. The background regions have been selected to be sufficiently far away such that the stellar light of the extended galaxies does not contaminate the spectra.

We do not require a Target Acquisition observation as our targets are point sources, and the default JWST pointing accuracy suffices with the MRS field of view. We will obtain simultaneous imaging to improve the astrometric calibration in data processing. The spectra for the TDE flares will be extracted from the data cube at the nuclear position, which is expected to be resolved from the background stellar light in the IFU data.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-7392	RA: 22 51 48.7580 (342.9531583d) Dec: -20 36 29.03 (-20.60806d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Spiral galaxies]				
(2)	MCG-01-07-028	RA: 02 40 23.7550 (40.0989792d) Dec: -02 43 41.73 (-2.72826d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Spiral galaxies]				
(3)	2MASX-J01464244+3230295	RA: 01 46 42.4463 (26.6768596d) Dec: +32 30 29.31 (32.50814d) Equinox: J2000	Proper Motion RA: -6.221478401047267E-5 sec of time/yr Proper Motion Dec: -3.119999746559188E-4 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Spiral galaxies]				
(4)	NGC-2981	RA: 09 44 56.5708 (146.2357117d) Dec: +31 05 52.31 (31.09786d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Spiral galaxies]				
(5)	NGC7392-BACKGROUND	RA: 22 51 43.5000 (342.9312500d) Dec: -20 36 51.80 (-20.61439d) Equinox: J2000		
Comments: Category=Calibration Description=[Telescope/sky background]				
(6)	MCG-01-07-028-BACKGROUND	RA: 02 40 24.0500 (40.1002083d) Dec: -02 44 8.60 (-2.73572d) Equinox: J2000		
Comments: Category=Calibration Description=[Telescope/sky background]				
(7)	KUG0143-BACKGROUND	RA: 01 46 40.9000 (26.6704167d) Dec: +32 30 14.70 (32.50408d) Equinox: J2000		
Comments: Category=Calibration Description=[Telescope/sky background]				
(8)	NGC2981-BACKGROUND	RA: 09 44 52.9000 (146.2204167d) Dec: +31 06 35.70 (31.10992d) Equinox: J2000		
Comments: Category=Calibration Description=[Telescope/sky background]				

Observation

Proposal 3696, Observation 1

Wed Nov 01 17:00:12 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Observation 2]

Diagnostics

(Observation 1) Warning (Form): Imager Filter overlap.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-7392	RA: 22 51 48.7580 (342.9531583d) Dec: -20 36 29.03 (-20.60806d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	60	1	1	Dither 1	4	4	666.01	
1	SHORT(A)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	
2		IMAGER	F1000W	FASTR1	60	1	1	Dither 1	4	4	666.01	
2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	
3		IMAGER	F1800W	FASTR1	90	1	1	Dither 1	4	4	999.014	
3	LONG(C)	MRSLONG		FASTR1	90	1	1	Dither 1	4	4	999.014	
3	LONG(C)	MRSSHORT		FASTR1	90	1	1	Dither 1	4	4	999.014	

Special Requirements	
	Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 3696, Observation 2

Wed Nov 01 17:00:12 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Observation 1]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	NGC7392-BACKGROUND	RA: 22 51 43.5000 (342.9312500d) Dec: -20 36 51.80 (-20.61439d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	60	1	1	None	1	1	166.502	
1	SHORT(A)	MRSLONG		FASTR1	60	1	1	None	1	1	166.502	
1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	None	1	1	166.502	
2		IMAGER	F1000W	FASTR1	60	1	1	None	1	1	166.502	
2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	None	1	1	166.502	
2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	None	1	1	166.502	
3		IMAGER	F1800W	FASTR1	90	1	1	None	1	1	249.754	
3	LONG(C)	MRSLONG		FASTR1	90	1	1	None	1	1	249.754	
3	LONG(C)	MRSSHORT		FASTR1	90	1	1	None	1	1	249.754	

Special Requirements	Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 3696, Observation 3

Wed Nov 01 17:00:12 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Observation 4]

Diagnostics

(Observation 3) Warning (Form): Imager Filter overlap.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	MCG-01-07-028	RA: 02 40 23.7550 (40.0989792d) Dec: -02 43 41.73 (-2.72826d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	80	1	1	Dither 1	4	4	888.013	
1	SHORT(A)	MRSLONG		FASTR1	80	1	1	Dither 1	4	4	888.013	
1	SHORT(A)	MRSSHORT		FASTR1	80	1	1	Dither 1	4	4	888.013	
2		IMAGER	F1000W	FASTR1	80	1	1	Dither 1	4	4	888.013	
2	MEDIUM(B)	MRSLONG		FASTR1	80	1	1	Dither 1	4	4	888.013	
2	MEDIUM(B)	MRSSHORT		FASTR1	80	1	1	Dither 1	4	4	888.013	
3		IMAGER	F1800W	FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 3696, Observation 4

Wed Nov 01 17:00:12 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Observation 3]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	MCG-01-07-028-BACKGROUND	RA: 02 40 24.0500 (40.1002083d) Dec: -02 44 8.60 (-2.73572d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	80	1	1	None	1	1	222.003	
1	SHORT(A)	MRSLONG		FASTR1	80	1	1	None	1	1	222.003	
1	SHORT(A)	MRSSHORT		FASTR1	80	1	1	None	1	1	222.003	
2		IMAGER	F1000W	FASTR1	80	1	1	None	1	1	222.003	
2	MEDIUM(B)	MRSLONG		FASTR1	80	1	1	None	1	1	222.003	
2	MEDIUM(B)	MRSSHORT		FASTR1	80	1	1	None	1	1	222.003	
3		IMAGER	F1800W	FASTR1	100	2	1	None	1	2	557.783	
3	LONG(C)	MRSLONG		FASTR1	100	2	1	None	1	2	557.783	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	None	1	2	557.783	

Special Requirements	
	Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 3696, Observation 5

Wed Nov 01 17:00:13 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Observation 6]

Diagnostics

(Observation 5) Warning (Form): Imager Filter overlap.

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	2MASX-J01464244+3230295	RA: 01 46 42.4463 (26.6768596d) Dec: +32 30 29.31 (32.50814d) Equinox: J2000	Proper Motion RA: -6.221478401047267E-5 sec of time/yr Proper Motion Dec: -3.119999746559188E-4 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	60	1	1	Dither 1	4	4	666.01	
1	SHORT(A)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	
2		IMAGER	F1130W	FASTR1	60	1	1	Dither 1	4	4	666.01	
2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	
3		IMAGER	F1800W	FASTR1	90	1	1	Dither 1	4	4	999.014	
3	LONG(C)	MRSLONG		FASTR1	90	1	1	Dither 1	4	4	999.014	
3	LONG(C)	MRSSHORT		FASTR1	90	1	1	Dither 1	4	4	999.014	

Special Requirements	
	Sequence Observations 5, 6, Non-interruptible

Observation

Proposal 3696, Observation 6

Wed Nov 01 17:00:13 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Observation 5]

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	KUG0143-BACKGROUND	RA: 01 46 40.9000 (26.6704167d) Dec: +32 30 14.70 (32.50408d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	60	1	1	None	1	1	166.502	
1	SHORT(A)	MRSLONG		FASTR1	60	1	1	None	1	1	166.502	
1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	None	1	1	166.502	
2		IMAGER	F1130W	FASTR1	60	1	1	None	1	1	166.502	
2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	None	1	1	166.502	
2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	None	1	1	166.502	
3		IMAGER	F1800W	FASTR1	90	1	1	None	1	1	249.754	
3	LONG(C)	MRSLONG		FASTR1	90	1	1	None	1	1	249.754	
3	LONG(C)	MRSSHORT		FASTR1	90	1	1	None	1	1	249.754	

Special Requirements	
	Sequence Observations 5, 6, Non-interruptible

Observation

Proposal 3696, Observation 7 Wed Nov 01 17:00:13 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Observation 8]

Diagnostics

(Observation 7) Warning (Form): Imager Filter overlap.

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	NGC-2981	RA: 09 44 56.5708 (146.2357117d) Dec: +31 05 52.31 (31.09786d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Spiral galaxies]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	SLOWR1	12	1	1	Dither 1	4	4	1146.716	
1	SHORT(A)	MRSLONG		SLOWR1	12	4	1	Dither 1	4	16	4873.544	
1	SHORT(A)	MRSSHORT		SLOWR1	12	4	1	Dither 1	4	16	4873.544	
2		IMAGER	F1130W	SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2	MEDIUM(B)	MRSLONG		SLOWR1	12	4	1	Dither 1	4	16	4873.544	
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	4	1	Dither 1	4	16	4873.544	
3		IMAGER	F1800W	SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3	LONG(C)	MRSLONG		SLOWR1	12	10	1	Dither 1	4	40	12327.199	
3	LONG(C)	MRSSHORT		SLOWR1	12	10	1	Dither 1	4	40	12327.199	

Special Requirements	
	Sequence Observations 7, 8, Non-interruptible

Observation

Proposal 3696, Observation 8
Wed Nov 01 17:00:13 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Observation 7]

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	NGC2981-BACKGROUND	RA: 09 44 52.9000 (146.2204167d) Dec: +31 06 35.70 (31.10992d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	SLOWR1	12	1	1	None	1	1	286.679	
1	SHORT(A)	MRSLONG		SLOWR1	12	4	1	None	1	4	1218.386	
1	SHORT(A)	MRSSHORT		SLOWR1	12	4	1	None	1	4	1218.386	
2		IMAGER	F1130W	SLOWR1	12	1	1	None	1	1	286.679	
2	MEDIUM(B)	MRSLONG		SLOWR1	12	4	1	None	1	4	1218.386	
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	4	1	None	1	4	1218.386	
3		IMAGER	F1800W	SLOWR1	12	1	1	None	1	1	286.679	
3	LONG(C)	MRSLONG		SLOWR1	12	10	1	None	1	10	3081.8	
3	LONG(C)	MRSSHORT		SLOWR1	12	10	1	None	1	10	3081.8	

Special Requirements	
	Sequence Observations 7, 8, Non-interruptible